

TMS, DMS and CMS Usage Guide for Falcon BMS 4.38.1

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Chapter 1

HOTAS Fundamentals

The F-16's HOTAS controls enable the pilot to execute critical flight, sensor, and weapon functions without removing hands from the throttle and stick grips. Within this architecture, three switches — the Display Management Switch (DMS), Target Management Switch (TMS), and Countermeasures Management Switch (CMS) — provide the primary interface for display management, sensor control, and defensive systems and form the core of the HOTAS architecture.

Effective use of these switches requires familiarity with two foundational concepts. The first is the Sensor of Interest (SOI), the mechanism that determines which display receives HOTAS inputs. The second is Master Modes (NAV, A-A, A-G), the operational contexts that govern available weapon systems and sensor configurations. This chapter establishes the conceptual framework necessary for the detailed switch behavior covered in Chapters ??, ??, and ??.

1.1 Sensor of Interest (SOI) — Concept and Architecture

The Sensor of Interest (SOI) is a designation that identifies which display currently receives HOTAS cursor slew commands and Target Management Switch (TMS) inputs. At any moment, only one display can hold SOI designation. This single-point control architecture enables the pilot to interact with multiple sensors and displays using HOTAS controls without manually selecting different switches for different sensors.

1.1.1 The SOI Concept

The F-16 cockpit contains three basic displays: the Head-Up Display (HUD) and two Multifunction Displays (MFD). The MFD can present various formats or pages, including radar, targeting pod, weapon management and tactical information. The pilot interacts with these displays through HOTAS controls, particularly the cursor slew control (CURSOR/ENABLE) and the Target Management Switch (TMS).

The Sensor of Interest (SOI) is the designation that identifies which display currently receives HOTAS inputs. At any moment, only one display holds SOI designation. When the pilot slews the cursor or presses TMS Up, the command is directed to whichever display is designated as SOI.

The SOI mechanism manages *where* HOTAS inputs are directed, not *what* those inputs do. The specific function of a HOTAS input — for example, whether TMS Up designates a target, breaks a track, or cycles through a mode — depends on the sensor mode, current Master Mode, and weapon selected. The SOI ensures that the input reaches the intended sensor; the sensor determines how the input is interpreted.

This design reflects the HOTAS philosophy: minimize pilot workload by consolidating control inputs through a limited number of switches while preserving the flexibility to interact with any sensor

as needed.

1.1.2 SOI Indicators and Visual Feedback

The pilot can immediately identify which display holds SOI designation through unambiguous visual indicators:

- **HUD as SOI:** An asterisk symbol (*) appears in the upper left corner of the HUD, above the airspeed scale.
- **MFD as SOI:** A border outline is drawn around the edges of the MFD display, forming a box that clearly distinguishes the SOI display from the non-SOI display.
- **NOT SOI indication:** When a display format is capable of being SOI but is not currently designated as such, the text **NOT SOI** may appear on the format. This text appears only on SOI-capable sensor formats, not on formats that are inherently unable to be SOI.

These visual cues allow the pilot to confirm at a glance which display is receiving HOTAS inputs. This immediate feedback is essential during dynamic operations, where the pilot may rapidly transition SOI between displays to manage multiple sensors or respond to tactical developments.

1.1.3 SOI-Capable Displays and Context Dependency

Not all displays can be designated as SOI, and the availability of SOI designation varies by master mode and MFD format.

The master mode determines which displays are eligible for SOI designation. In some operational contexts, the HUD can be designated as SOI, allowing the pilot to manage HUD symbology through HOTAS inputs. In other contexts, particularly air-to-air engagements, SOI designation is restricted to MFD formats only. This restriction ensures that primary sensor displays remain the focus of HOTAS control during time-critical tactical situations.

Similarly, not all MFD formats are SOI-capable. Sensor formats that accept targeting and cursor control inputs—such as radar, targeting pod, and weapon displays—can be designated as SOI. Informational or configuration formats that do not require direct sensor interaction cannot be designated as SOI. This design aligns SOI availability with operational function: only displays that benefit from HOTAS cursor and targeting control are SOI-capable.

Understanding these context dependencies is essential for effective HOTAS use. The pilot must recognize not only how to designate a display as SOI, but also whether such designation is permitted in the current operational context.

For complete operational reference tables showing SOI availability by master mode and the full list of valid and invalid MFD formats, see Section ??.

1.2 Master Modes and Context-Sensitive Behavior

Section 1.1 established that SOI designation varies by master mode—some master modes permit HUD as SOI, while others restrict SOI to MFD formats only. This variation reflects a broader principle: master mode is the highest-level configuration state of the F-16 avionics system, determining not only SOI availability but also which HOTAS functions are active and how switch inputs are interpreted.

Each master mode pre-configures the cockpit displays, sensor modes, and weapon options for specific missions and purposes. The master mode determines which displays can be designated as SOI, which HOTAS functions are available, and how individual switch inputs are interpreted by the avionics suite. Understanding master modes is essential to understanding why HOTAS switches behave differently in different operational contexts.

1.2.1 Master Mode Overview

The F-16 provides three primary Master Modes (NAV, A-A, and A-G) and two override modes (DGFT and MSL OVRD), each optimized for a specific phase of flight or mission objective:

- **Navigation (NAV):** The default master mode, used for transit, fuel management, and general situational awareness. In NAV, the avionics system prioritizes navigation displays and provides access to both air-to-air and air-to-ground sensors without committing to a specific weapon employment mode.
- **Air-to-Air (A-A):** Configures the avionics suite for beyond-visual-range (BVR) and within-visual-range (WVR) air-to-air engagements. A-A mode enables radar search and track modes, datalink integration, and air-to-air weapon selection. HUD symbology shifts to air-to-air cues, and the MFD formats default to radar and tactical displays.
- **Air-to-Ground (A-G):** Optimizes the cockpit for air-to-ground weapon delivery. A-G mode provides access to ground attack sensors (radar ground modes, targeting pod, HARM attack display), weapon delivery submodes (CCIP, CCRP, DTOS, etc.), and stores management functions. The HUD displays air-to-ground weapon delivery symbology.
- **Dogfight (DGFT) and Missile Override (MSL OVRD):** These are override modes accessed via a throttle-mounted switch. Both are optimized for air-to-air engagements, similar to the A-A Master Mode, but provide immediate access to specific weapon configurations. DGFT configures the cockpit for close-range visual combat, typically defaulting to a high-off-boresight radar mode (ACM) and short-range missile (AIM-9) selection. MSL OVRD provides rapid access to medium-range missile employment (AIM-120) with optimized radar modes. Both override modes take precedence over any other master mode except Emergency Jettison.

Each master mode establishes default configurations for sensors, displays, and weapons. These defaults can be pre-programmed via the Data Transfer Cartridge (DTC) during mission planning or adjusted manually during flight, as detailed in Section ???. The master mode determines not only what the pilot sees on the displays, but also how HOTAS switches function. A single HOTAS input — such as TMS Up — produces different results in NAV, A-A, and A-G modes because the underlying sensor and weapon context has changed. For complete master mode details, see Dash-34 § 2.1.1.2.1

1.2.2 Why HOTAS Switches Behave Differently by Master Mode

HOTAS switches behave differently by master mode because the master mode determines the operational context. A single physical switch action—such as TMS Up—must support different tactical functions depending on whether the pilot is navigating, engaging an air target, or delivering ordnance to ground. Rather than require separate switches for each function, the F-16 design uses master

mode to reinterpret the same switch input based on the current mission phase. This context-sensitive architecture enables a limited number of HOTAS controls to perform a wide range of mission-critical tasks.

The pilot must be aware of the current master mode to predict HOTAS behavior. This awareness is not an additional cognitive burden—the master mode is a natural reflection of the mission phase. The principle that master mode determines context, and context determines function, is the organizing principle behind the detailed HOTAS tables and procedures in Chapters ??, ??, and ?. Those chapters document the specific functions of DMS, TMS, and CMS in various master modes, sensor modes, and weapon configurations.

1.3 The Three Switches: DMS, TMS, and CMS

The Display Management Switch (DMS), Target Management Switch (TMS), and Countermeasures Management Switch (CMS) form the core of HOTAS sensor and display management in the F-16. Each switch serves a distinct function, and understanding their roles in relation to SOI and master modes is essential before examining the detailed behaviors covered in subsequent chapters.

1.3.1 Display Management Switch (DMS) — Overview

The Display Management Switch (DMS) is a four-way, spring-loaded hat switch located on the flight stick. Its primary functions are SOI selection and MFD format cycling. The DMS does not perform tactical actions such as target designation or weapon release; instead, it manages *which display or sensor* receives HOTAS inputs from other controls.

DMS Up attempts to designate the HUD as the Sensor of Interest, provided the current master mode permits HUD SOI designation (NAV and A-G allow this; A-A, DGFT, and MSL OVRD do not). DMS Down cycles the SOI designation between the left and right MFDs, or transitions SOI from the HUD back to an MFD. DMS Left and DMS Right cycle through the MFD format pages displayed on the left and right MFDs, respectively, independently of which display is currently SOI.

The DMS is a transversal SOI manager. It controls the routing of HOTAS inputs to displays, but does not execute sensor-specific or weapon-specific functions. This distinction separates the DMS from the TMS, which operates *through* the SOI to execute tactical actions such as target designation, track management, and weapon cueing.

Detailed DMS functionality, including SOI selection mechanics, MFD format cycling behavior, and master mode constraints, is covered in Chapter ??.

1.3.2 Target Management Switch (TMS) — Overview

The Target Management Switch (TMS) is a four-way, spring-loaded hat switch located on the throttle grip. Its primary function is target management and designation, controlling how the pilot interacts with radar contacts, targeting pod tracks, weapon employment cues, and other sensor-derived targeting information.

Unlike the DMS, which manages *which* display is active, the TMS operates *through* the currently active SOI to execute tactical functions. The specific behavior of each TMS direction depends on the master mode, the sensor mode, and which display holds the SOI designation. For example, when the Fire Control Radar (FCR) is SOI in Air-to-Air mode, TMS Up commands a target designation, locking

the radar onto a search contact. When the Targeting Pod (TGP) is SOI in Air-to-Ground mode, TMS Up may break an existing track, establish a cursor zero point, or transition to a post-designate mode, depending on the A-G submode and weapon selected.

The TMS is fundamentally context-dependent. The pilot must understand not only the current master mode, but also which display is SOI and what sensor mode that display is operating in. This context-sensitivity allows a single four-way switch to perform dozens of distinct functions across the range of F-16 mission profiles.

Detailed TMS functionality, including target designation procedures, sensor-specific behaviors, and master mode contexts, is covered in Chapter ??.

1.3.3 Countermeasures Management Switch (CMS) — Overview

The Countermeasures Management Switch (CMS) is a four-way, spring-loaded hat switch located on the throttle grip. Its primary function is countermeasures dispensing, controlling the release of chaff and flare from the ALE-47 Countermeasures Dispenser System (CMDS) and managing Electronic Countermeasures (ECM) pod transmissions.

Unlike the DMS and TMS, the CMS operates largely independently of Sensor of Interest designation. CMS commands interact directly with defensive systems — the CMDS, ECM pod, and Radar Warning Receiver (RWR) — rather than routing through the SOI architecture. The pilot actuates the CMS in response to threat warnings displayed on the RWR, and the countermeasures system responds according to pre-programmed or manually configured dispense patterns.

CMS Up, CMS Down, CMS Forward, and CMS Aft each trigger specific countermeasures responses. The exact chaff and flare quantities dispensed, and the enabling or disabling of ECM transmissions, depend on the CMDS programming and the CMS manual mode settings (if selected). The CMS behavior is relatively consistent across master modes, although certain operational contexts (such as air-to-ground attack profiles) may favor specific countermeasures programs.

The CMS provides the pilot with immediate, hands-on control of defensive systems without requiring menu navigation or display interaction. This direct-access design reflects the time-critical nature of countermeasures employment in response to surface-to-air or air-to-air threats.

Detailed CMS functionality, including CMDS program interaction, ECM pod control, and manual mode operation, is covered in Chapter ??.